

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) April 7th, 2012

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Let X_1, X_2 be independent with density $f_X(x; \beta) = \frac{e^{-x/\beta} x}{\beta^2}$, $x > 0$.

- a) Find the MLE of β .
- b) Find the MLE of $e^{\beta+1}$.

2. An urn initially contains 1 red ball and 1 blue ball. At each stage a ball is randomly withdrawn and replaced by two other balls of the same color. (For instance, if the red ball is initially chosen, then there would be 2 red and 1 blue ball in the urn when the next selection occurs.) Let $P_{n,i}$ be the probability that there are exactly i red balls in the urn after n stages have been completed $1 \leq i \leq n+1$.

- a) Derive an equation for $P_{n,i}$ in terms of $P_{n-1,j}$, with boundary conditions.
- b) Calculate $P_{9,1}$ and $P_{9,2}$.

3. Let X be a single observation from a population whose density is

$$f_X(x; 0) = 0x^{0-1}, \quad 0 < x < 1 \\ = 0, \quad \text{otherwise}$$

Find the most powerful test of size $\alpha = .05$ for $H_0: 0 = 1$ vs $H_1: 0 = 2$

4. There are two bins that contain two different types of batteries. When in use, batteries in bin i last (in hours) an exponentially distributed time with rate $\lambda_i, i = 1, 2$. Three batteries are independently chosen from the bins, each with probability p_i from bin $i, p_1 + p_2 = 1$. If all batteries are still operating after t hours of use, what is the probability that all three will still be operating after an additional s hours?